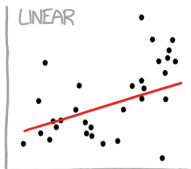


# Towards Modern Regression

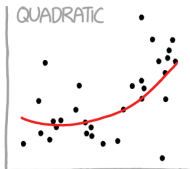
Daniel Lawson University of Bristol

Lecture 02.1 (v4.0.0)

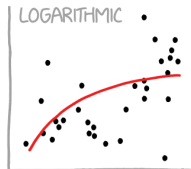
# CURVE-FITTING METHODS AND THE MESSAGES THEY SEND



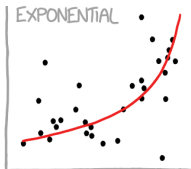
"HEY, I DID A  
REGRESSION."



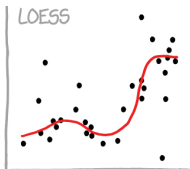
"I WANTED A CURVED  
LINE, SO I MADE ONE  
WITH MATH."



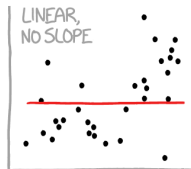
"LOOK, IT'S  
TAPERING OFF!"



"LOOK, IT'S GROWING  
UNCONTROLLABLY!"

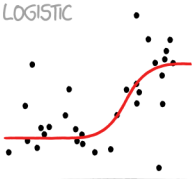


"I'M SOPHISTICATED, NOT  
LIKE THOSE BUMBLING  
POLYNOMIAL PEOPLE."



"I'M MAKING A  
SCATTER PLOT BUT  
I DON'T WANT TO."

LOGISTIC



"I NEED TO CONNECT THESE TWO LINES, BUT MY FIRST IDEA DIDN'T HAVE ENOUGH MATH."

CONFIDENCE INTERVAL



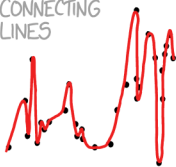
"LISTEN, SCIENCE IS HARD. BUT I'M A SERIOUS PERSON DOING MY BEST."

PIECEWISE



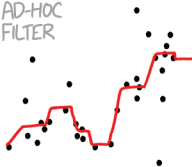
"I HAVE A THEORY, AND THIS IS THE ONLY DATA I COULD FIND."

CONNECTING LINES



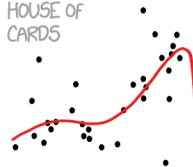
"I CLICKED 'SMOOTH LINES' IN EXCEL."

AD-HOC FILTER



"I HAD AN IDEA FOR HOW TO CLEAN UP THE DATA. WHAT DO YOU THINK?"

HOUSE OF CARDS



"AS YOU CAN SEE, THIS MODEL SMOOTHLY FITS THE- WAIT NO NO DON'T EXTEND IT AAAAAA!!!"

# Signposting

- ▶ This lecture covers:
  - ▶ Classical regression
  - ▶ Towards Modern Regression - the vectorised version, which uses Matrix algebra.
  - ▶ Leave-one-out Cross Validation
- ▶ The maths here underpins almost all modern data science.

# Questions

- ▶ What is Regression (not) for?
- ▶ Can you relate this to advanced Machine Learning?
- ▶ When does even linear regression cost too much?
  - ▶ What can be done when we have too much data?
  - ▶ How does AI solve the problem?
- ▶ How do you know one model is “better” than another?

## Before we start: Vector Notation

- ▶ There are several choices of convention that we have to make
- ▶ Vectors of length  $k$  are also matrices, but are they  $k \times 1$  or  $1 \times k$ ?
- ▶ We use  $k \times 1$ , i.e. column vectors
- ▶ Similarly there are choices about matrix derivatives
- ▶ We use derivative with respect to a column vector as a row vector
- ▶ Some resources differ and have everything transposed as a consequence

# Regression

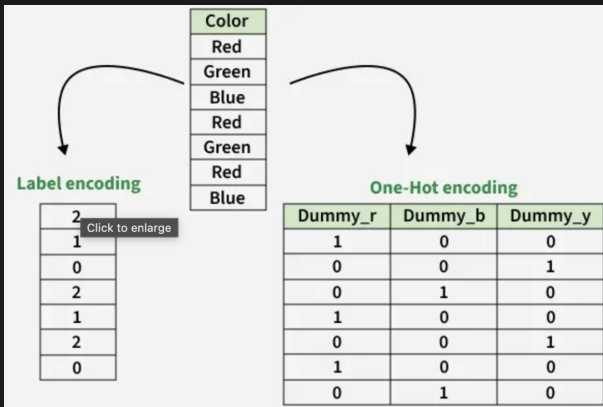
- ▶ *A reminder: understanding univariate regression is a prerequisite*
- ▶ **Regression**, considers the relationship of a response variable as determined by one or more explanatory variables.
  - ▶ Regression is designed to help **make predictions** of  $y$  when we observe  $x$ .
  - ▶ It is a **conditional model**, and **not** a joint model of  $x$  and  $y$ . This is its **strength**.
  - ▶ It predicts the **best guess** in **squared error loss**.
  - ▶ There is a probabilistic interpretation based on Normal Distributions.

# (Not) Causality

- ▶ Regression is often used as a tool to examine causality. . .
  - ▶ A and B share a causal relationship if a regression for B given A has an association, conditional on (“controlling for”) C (C=**everything else**)
  - ▶ This does not resolve whether A causes B, or B causes A
  - ▶ Since we don’t measure **everything else**, regression rarely establishes causality!
  - ▶ Further assumptions are needed to make a causal connection. This is known as **causal inference**.

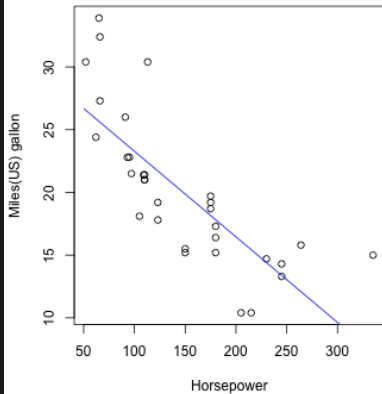
# Discrete predictors

- ▶ One Hot Encoding: each unique value or **level** is a binary predictor.
- ▶ Label Encoding: Convert categories to labels (“dictionary” of “tokens”)

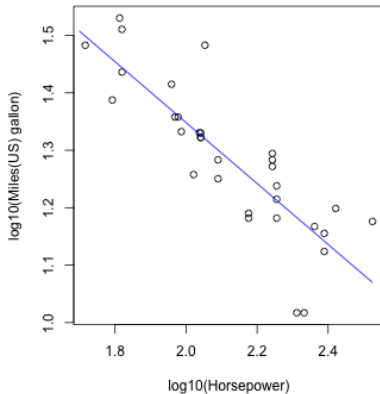


# Regression example

a) Scatter plot (mtcars MPG~HP)



b) Log-scale Scatter plot (mtcars MPG~HP)



## Multiple Regression example

```
> lm(mpg ~ cyl + hp + wt, data=mtcars) %>% summary
```

Call:

```
lm(formula = mpg ~ cyl + hp + wt, data = mtcars)
```

Residuals:

|  | Min     | 1Q      | Median  | 3Q     | Max    |
|--|---------|---------|---------|--------|--------|
|  | -3.9290 | -1.5598 | -0.5311 | 1.1850 | 5.8986 |

Coefficients:

|             | Estimate | Std. Error | t value | Pr(> t ) |     |
|-------------|----------|------------|---------|----------|-----|
| (Intercept) | 38.75179 | 1.78686    | 21.687  | < 2e-16  | *** |
| cyl         | -0.94162 | 0.55092    | -1.709  | 0.098480 | .   |
| hp          | -0.01804 | 0.01188    | -1.519  | 0.140015 |     |
| wt          | -3.16697 | 0.74058    | -4.276  | 0.000199 | *** |
| ---         |          |            |         |          |     |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

# Important measures of regression

- ▶ **R squared** (and adjusted R squared): variance explained/total variance. This tells us how predictable  $y$  is.
- ▶ The coefficients  $\beta_i$ .
  - ▶ These should be compared to their error  $\hat{\sigma}_i$ .
  - ▶ The ratio is a t-value ( $t_i = \beta_i/\hat{\sigma}_i$ ) from which a p-value can be calculated.
- ▶ F statistic and F test p-value:
  - ▶  $F$  is the ratio of the explained to unexplained variance, accounting for the **degrees of freedom**.
  - ▶ The full model compared to a null in which there are no explanatory variables.
  - ▶ Used in variable selection, ANOVA, etc.

# Regression is analogous to linear algebra with noise

- ▶ Most problems in Linear Algebra can be seen as **solving a system of linear equations**:

$$XA + \mathbf{b} = 0.$$

- ▶ Where  $X$  is an  $n$  by  $p$  matrix of data,
  - ▶  $A$  is an  $p$  by 1 matrix of coefficients,
  - ▶ and  $-\mathbf{b}$  is a  $n$ -vector of target values.
- ▶ However, in the presence of noise we seek the least-bad fit:

$$\operatorname{argmin}_{(A, \mathbf{b})} \|XA + \mathbf{b}\|_2^2 = \sum_{i=1}^N (\mathbf{x}_i A + b_i)^2$$

- ▶ i.e. we find  $A$  and  $\mathbf{b}$  such that they minimise the distance (in the squared  $L_2$  norm)
- ▶ Linear algebra solves this very effectively!

# Matrix form of least squares

- ▶ Consider data  $X'$  with  $p'$  **features** (columns) and  $n$  observations.
- ▶ Given the regression problem:

$$\mathbf{y} = X' \beta' + \mathbf{b} + \mathbf{e}$$

- ▶ to find:
  - ▶  $\beta'$  (a matrix dimension  $p' \times 1$ )
  - ▶ and  $b$ ,
  - ▶ to minimise 'error': in  $\sum_{i=1}^n e_i^2$

# Matrix form of least squares

- ▶ We construct a simpler representation by adding a constant feature:

$$X = \begin{bmatrix} 1 & X_{11} & \cdots & X_{1p'} \\ & & \cdots & \\ 1 & X_{n1} & \cdots & X_{np'} \end{bmatrix}$$

- ▶ which has  $p = p' + 1$  features.
- ▶ We now solve the analogous equation:

$$\mathbf{y} = X\beta + \mathbf{e}$$

- ▶ which has the same solution but is in a more convenient form.

# Mean Squared Error (MSE)

- ▶ The **prediction error** is:

$$\mathbf{e}(\beta) = \mathbf{y} - \mathbf{X}\beta$$

- ▶ Using the notation that  $\mathbf{e}$  is a  $n$  by 1 matrix
- ▶ The **estimation error** is written in matrix form:

$$\text{MSE}(\beta) = \frac{1}{n} \mathbf{e}^T \mathbf{e}$$

- ▶ Why?  $\mathbf{e}^T \mathbf{e} = \sum_{i=1}^n e_i^2$ 
  - ▶ Hence  $\text{MSE}(\beta)$  is a  $1 \times 1$  matrix, i.e. a scalar, and  $|\text{MSE}(\beta)| = \text{MSE}(\beta)$ .
  - ▶ Noticing this sort of thing makes the matrix algebra easier.
- ▶ We want to minimise this MSE with respect to the parameters  $\beta$ .

# How to do the Matrix Algebra

Lecture 13 of Cosma Shalizi's notes is a really helpful reminder!

- ▶ Look at the [Matrix Algebra Cheat Sheet](#) - specifically:
  - ▶ How does a transpose work?
  - ▶ How do you re-order elements?
  - ▶ How does a gradient work in linear and quadratic forms?

# Minimising MSE

- ▶ Taking (vector) derivatives with respect to  $\beta$ :

$$\nabla \text{MSE}(\beta) = \frac{1}{n}(\nabla \mathbf{y}^T \mathbf{y} - 2\nabla \beta^T \mathbf{X}^T \mathbf{y} + \nabla \beta^T \mathbf{X}^T \mathbf{X} \beta)$$

- ▶ Simplifying:

$$\frac{1}{n}(0 - 2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \beta)$$

- ▶ which is zero at the optimum  $\hat{\beta}$ :

$$\mathbf{X}^T \mathbf{X} \hat{\beta} - \mathbf{X}^T \mathbf{y} = 0$$

- ▶ with the solution:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.$$

- ▶ Exercise: For the case  $p' = 1$ , check that this solution is the same as you can find in regular linear algebra textbooks.

# Implications

- ▶ These vector calculations are repeated in many machine-learning methods
  - ▶ The details change but the principle remains
- ▶ Linear-Algebra loss minimisation techniques are extremely important
  - ▶ They often sit inside a wider argument, e.g. updated conditional on some other parameters

# Reflection

- ▶ By the end of the course, you should:
  - ▶ Be able to define **correlation** and **regression** in multivariate context
  - ▶ Be able to perform basic calculations using these concepts
  - ▶ Be able to extend intuition about their application.
  - ▶ Be able to follow the reasoning in a paper where things get complicated.
- ▶ Matrix algebra is worth reading up on!
  - ▶ Describe it for example in your assessments' reflection.

# Signposting

- ▶ Make sure to look at **02.1-Regression.R**
- ▶ The mathematics behind Modern Regression is analogous to the mathematics underpinning scalable Machine Learning. **It is very important.**
- ▶ For accessible material see **Cosma Shalizi's Modern Regression Lectures** (Lectures 13-14)
- ▶ Further reading in chapters 2.3 and 3.2 of **The Elements of Statistical Learning: Data Mining, Inference, and Prediction** (Friedman, Hastie and Tibshirani)
- ▶ Next up: 2.2 Statistical Testing

## Appendix: Expectations

- ▶ If the data were generated by our model(!) then they are described by an RV  $\mathbf{Y}$  (an  $n$ -vector):

$$\mathbf{Y}_i = \mathbf{x}_i\beta + \mathbf{e}_i$$

- ▶  $\mathbf{x}_i$  is still a vector but *not* a Random Variable!
- ▶  $\mathbf{e}$  is an  $n \times 1$  matrix of RVs with mean  $\mathbf{0}$  and covariance  $\sigma_s^2 \mathbf{I}$ .
- ▶ From this it is straightforward to show that the **fitted values are unbiased**:

$$\mathbb{E}[\hat{\mathbf{y}}] = \mathbb{E}[\mathbf{H}\mathbf{Y}] = \mathbf{x}\beta$$

- ▶ using the properties of Expectations with the symmetry and idempotency of  $\mathbf{H}$ .

# Covariance

- ▶ *A reminder: understanding covariance and correlation is a prerequisite*
- ▶ covariance is simply the second (central) moment:

$$\text{cov}(X, Y) = \mathbb{E} [(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])]$$

- ▶ Recall that we typically use **unbiased** estimators which often slightly differ from the natural theoretical analogue. The **sample covariance** is:

$$\text{cov}(X, Y) = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})(Y_i - \bar{Y})$$

# Linear algebra view of covariance

- ▶ The **covariance matrix** of a random variable  $X$
- ▶ Where  $X$  is a vector-valued RV with length  $k$ ,
- ▶ has entries:

$$\text{Cov}(X)_{ij} = \text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)].$$

- ▶ The matrix form for this is:

$$\Sigma = \mathbb{E}[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^T],$$

- ▶ Where  $\mu = \mathbb{E}[X]$ .

# Correlation

- ▶ Correlation is simply a **normalised measure** of covariance.

$$\rho_{X,Y} = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

- ▶ It takes values between -1 and 1.
- ▶ Sample correlation uses the unbiased estimator of covariance, to account for the number of degrees of freedom in the data.
- ▶ **Question:** What should we (not) take the correlation of?
  - ▶ See rank correlation, canonical correlation, etc.

# Linear algebra view of correlation

- ▶ Division by standard deviations is required to correctly generalise the **scalar correlation**:

$$\text{Corr}(X, Y) = \frac{\mathbb{E}[(X - \mu_X)(Y - \mu_Y)]}{\sigma_X \sigma_Y}.$$

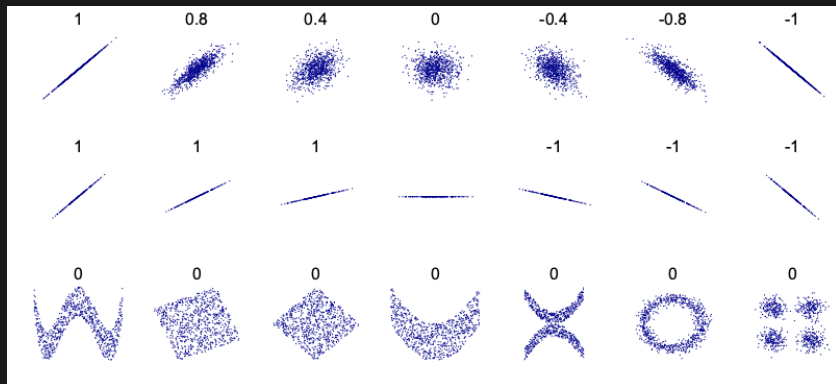
- ▶ The **matrix form** for correlation is:

$$\text{Corr}(X) = (\text{diag}(\Sigma))^{-1/2} \Sigma (\text{diag}(\Sigma))^{-1/2}$$

- ▶ The matrix inversion is not computationally challenging because it is for a **diagonal matrix**.

# Examples

From [Wikipedia: Correlation\\_and\\_dependence](#)



## Appendix: Covariance

- ▶ Similarly, it is straightforward to show that

$$\text{Var}[\hat{\mathbf{y}}] = \sigma_s^2 \mathbf{H}$$

using the properties of Variances with the symmetry and idempotency of  $\mathbf{H}$ .

- ▶ In other words, the covariance of the fitted values is determined entirely by the structure of the covariates, via the Hat matrix.

## Appendix: Quantifying Out-of-sample prediction accuracy

- ▶ ▶ For the second term in  $E(e_i'^2) = \mathbb{E}(e_i^2) + 2\text{Cov}[y_i, \hat{y}_i]$ ,
- ▶ We're now able to compute the covariance between  $y_i$  and its prediction  $\hat{y}_i$ :

$$\text{Cov}[y_i, \hat{y}_i] = \sigma^2 H_{ii}$$

- ▶ The mean out-of-sample prediction error is

$$\mathbb{E}(e'^2) = n^{-1} \sum_{i=1}^n e_i'^2 = n^{-1} \sum_{i=1}^n e_i^2 + 2n^{-1} \text{tr}(H)$$

- ▶ We show in Worksheet 2.2A that  $\text{tr}(H) = \sigma^2 p$  where  $p$ =number of predictors.
- ▶ The **optimism** is defined as  $2n^{-1}\sigma^2 p$ .
- ▶ The optimism grows with  $\sigma^2$  and  $p$  but shrinks with  $n$ . It is used to define the **model selection criteria**  $\Delta C_p$  which is minimised:

$$\Delta C_p = MSE_1 - MSE_2 + \frac{2}{n} \hat{\sigma}^2 (p_1 - p_2)$$